

Muhammad Talha Zaheer

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PROFESSIONAL SUMMARY

Computer scientist specializing in Agentic AI and secure software architecture. Experienced in building autonomous AI solutions with Next.js, Node.js, and the Vercel AI SDK—including tool calling, streaming generation, and production deployment. Combines full-stack development with a defensive security mindset shaped by hands-on cybersecurity training and instruction. Passionate about technical education, community leadership, and designing intelligent systems that are both innovative and secure.

TECHNICAL SKILLS

Languages	Python, JavaScript, TypeScript, Bash, SQL
Frameworks	Next.js, React, Node.js
AI & ML	Agentic AI, LLMs, Vercel AI SDK, Tool Calling, Pipecat AI, Deepgram, Cartesia, Daily PSTN, Prompt Optimization, LM Studio
DevOps & Deployment	Git/GitHub, GitHub Actions, Docker, CI/CD, Vercel, Railway, Digital Ocean, Supabase
Tools & IDEs	Postman, Cursor, VS Code, Google Antigravity, VirtualBox
Project Management	Monday.com, Linear

PROFESSIONAL EXPERIENCE

Core Edge Solutions

AI Developer

May 2025 – May 2026

Lahore, Pakistan

- Built production voice agents using Retell and Pipecat AI, integrating Deepgram (STT), OpenAI (LLM), and Cartesia (TTS) for cost-efficient conversational calling.
- Implemented tool calling for call transfer, call end, voicemail detection, and external API integration; optimized Pacifica Bot prompts after learning TMS domain workflows.
- Developed *Zuum* (FastAPI + React), an internal platform that outbound-calls dispatch drivers for shipment status updates.
- Designed GitHub Actions CI/CD pipelines to automate deployments on Railway.
- Built Puppeteer automations for Trinium TMS: adding container notes, updating empty container status, and processing acknowledged containers.
- Wrote SQL queries against Trinium TMS, exposed data through custom APIs, and compiled reports for container tracking.

Cyber Security Society – UMT

Cybersecurity Mentor

Feb 2025 – Jul 2025

Lahore, Pakistan

- Led weekly workshops on network security, threat modeling, and defensive security practices.
- Ran hands-on labs with Nmap, Wireshark, and Burp Suite to connect theory to real scenarios.
- Mentored students and grew an active campus security community focused on careers in infosec.

Professional Freelancing Training Program

Cybersecurity Intern

Nov 2024 – Apr 2025

Lahore, Pakistan

- Conducted web application penetration testing for SQL injection, file inclusion, and unrestricted file upload vulnerabilities.
- Performed network reconnaissance with Nmap and analyzed HTTP/traffic flows using Burp Suite and Wireshark.
- Researched and lab-tested exploits including EternalBlue (MS17-010), macro-based attacks, and GHDB-

- driven reconnaissance; tracked emerging CVEs for early evaluation.
- Configured WAF rules and pfSense firewalls to protect web applications; practiced digital forensics on hard drives and disk images.

Freelance

Full Stack Developer

Mar 2024 – Oct 2024

Remote

- Led a small team to build and deploy Electron.js POS systems for restaurant clients from open-project requirements through production rollout.
- Architected offline-first storage with a local database and Supabase cloud sync for periodic multi-device data synchronization.
- Integrated multiple thermal printer connections for receipt and kitchen order printing across deployed POS installations.

RELEVANT PROJECTS

Voice Agent

AI Agent

Built production voice agents with Pipecat AI, Deepgram STT, OpenAI, and Cartesia TTS; implemented tool calling for transfers, voicemail detection, and API integrations.

POS System

Professional Work

Electron.js point-of-sale system for restaurants with order management, multi-printer support, and production deployment.

Waiter App

Professional Work

Staff-facing application for taking and syncing restaurant orders with the POS backend across service workflows.

CTF Platform

Web Application

Web platform for hosting and managing Capture The Flag cybersecurity competitions with practical challenge workflows.

File Sync System

C++ Application

C++ file synchronization utility for reliable local file management and structured data transfer.

Portfolio: talhazaheer.dev/#projects

EDUCATION

University of Management and Technology (UMT)

Bachelor of Science, Computer Science

Nov 2019 – Aug 2025

Lahore, Pakistan

Professional Freelancing Training Program (NIAIS)

Diploma, Cyber Security

Jun 2024 – May 2025

Lahore, Pakistan

NAVTTTC

Certification, Digital Forensic & Cyber Security

Apr 2025 – Jul 2025

Pakistan

ADDITIONAL EXPERIENCE

Hunarmand Punjab

Course Instructor

Sep 2025 – Oct 2025

Punjab, Pakistan

Delivered structured technical training as part of a government skills development initiative.